Git hub:

1) Version controlling
2) Deployment
3) Collaboration

To know whether git is present in your system or not:

```
C:\Users\karthikeya>git --version
git version 2.52.0.windows.1
```

.gitignore filename is a file that tells Git to ignore certain files when pushing code to GitHub.

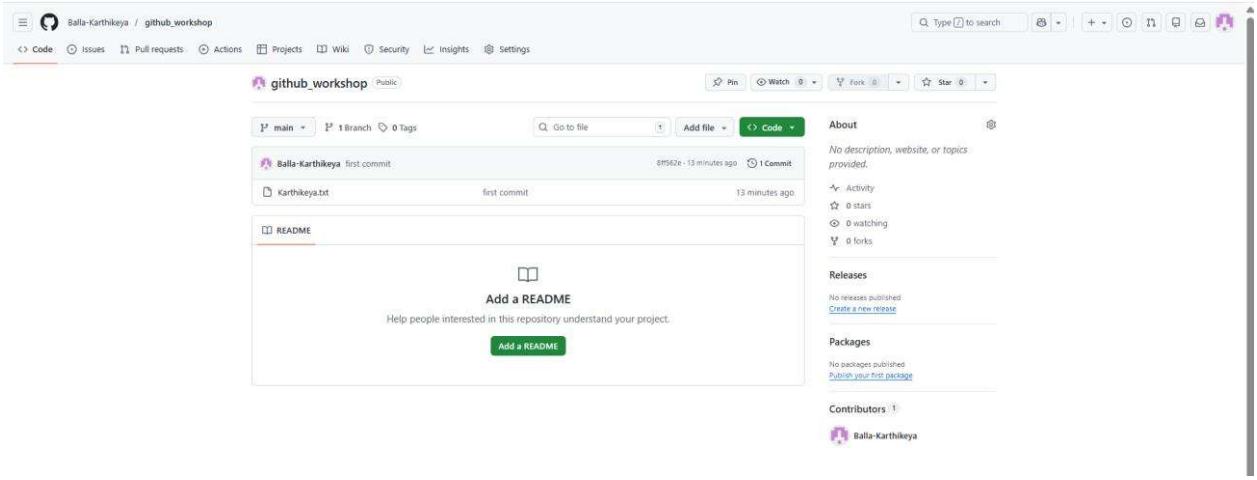
A GitHub Repository (Repo) is a place where your project code and files are stored online on GitHub

Git: offline

GitHub: online

Connection between your system and git hub and push

```
C:\Users\karthikeya>cd Desktop

C:\Users\karthikeya\Desktop>cd git_projects

C:\Users\karthikeya\Desktop\git_projects>git config --global user.email "karthikeyaballa123@gmail.com"

C:\Users\karthikeya\Desktop\git_projects>git config --global user.name "Balla-Karthikeya"

C:\Users\karthikeya\Desktop\git_projects>git init
Initialized empty Git repository in C:/Users/karthikeya/Desktop/git_projects/.git/

C:\Users\karthikeya\Desktop\git_projects>git add .

C:\Users\karthikeya\Desktop\git_projects>git commit -m "first commit"
[master (root-commit) 8ff562e] first commit
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 Karthikeya.txt

C:\Users\karthikeya\Desktop\git_projects>git branch -M main

C:\Users\karthikeya\Desktop\git_projects>

C:\Users\karthikeya\Desktop\git_projects>git remote add origin https://github.com/Balla-Karthikeya/github_workshop.git

C:\Users\karthikeya\Desktop\git_projects>git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 225 bytes | 112.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Balla-Karthikeya/github_workshop.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

C:\Users\karthikeya\Desktop\git_projects>
```



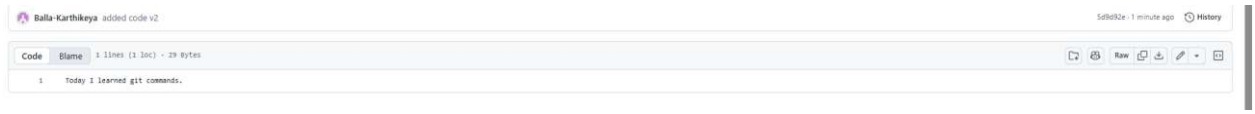
“How can I send new changes in a file to GitHub when the file already exists in the repository?”

changes modification:

Whenever we make a change in GitHub specific repository which is already connected with our local files.

```
C:\Users\karthikeya\Desktop\git_projects>git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   Karthikeya.txt

no changes added to commit (use "git add" and/or "git commit -a")
```

```
no changes added to commit (use "git add" and/or "git commit -a")

C:\Users\karthikeya\Desktop\git_projects>git add .

C:\Users\karthikeya\Desktop\git_projects>git commit -m "added code v2"
[main 5d9d92e] added code v2
 1 file changed, 1 insertion(+)

C:\Users\karthikeya\Desktop\git_projects>git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Writing objects: 100% (3/3), 287 bytes | 287.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Balla-Karthikeya/github_workshop.git
   8ff562e..5d9d92e  main -> main

C:\Users\karthikeya\Desktop\git_projects>
```



Git collaboration:

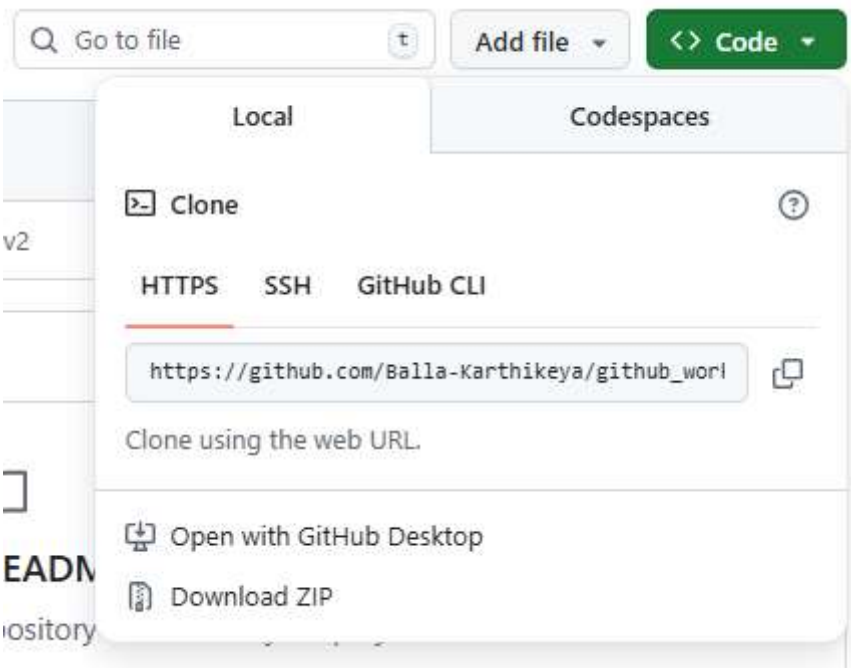
To access a particular repository by multiple employees we are going to collaborate with him or her

1) Go to settings
2) Select collaborators in left side
3) Send invitation using email or username
4) The other person will accept the invitation

To access the repository code: we need to clone that specific repository using git clone followed by URL

Clone

git clone https://github.com/Balla-Karthikeya/github_workshop.git



After cloning we need to create a new branch

To switch to multiple branches, use git switch command followed by branch name

Using git status, we can find in which we are working.

Using git branch we can find list of branches we are having

git status -> main

git checkout –b name (Krishna)

git status -> Krishna

git switch main -> main

git switch Krishna -> Krishna

git branch

Changes made by another person

Once we successfully created the branch, we are going to add new changes by switching into another branch

untracked

git add .

git commit –m “message”

git push -u origin branch_name

Change/modifications which we need to get from the main branch into to our local system

git fetch: which brings all the branches which are created

git merge origin/branch name: which will get the changes into our system